ENGINEERING RFC

RFC-001 · Real-Data Strategy

Sourcing live and historical tournament data

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Portfolio artifact — based on a real, production-grade system designed and built by the author.

Contents

1. Metadata

Field	Value
RFC	001
Title	Real-data strategy for live and historical tournaments
Author	Tobi Smith-Kayode
Status	Accepted
Decision	Adopt a hybrid: StatsBomb open data (historical) + API-Football (live 2026)
Affects	Data adapters, the normalized snapshot, caching, cost model

2. Summary

The platform needs **real** football data for two distinct jobs: a **live** feed for the in-progress 2026 World Cup, and **historical** event data for past tournaments. No single affordable source does both well. This RFC evaluates the options and recommends a **hybrid**: StatsBomb open data for history, API-Football for live — normalized behind one adapter interface so the rest of the system is unaware of the split.

3. Context & Problem

The product's core promise is letting people play with 2026 analytics as it happens, with historical tournaments available for context using the same tools. That promise imposes hard requirements on the data layer:

- **Live coverage of 2026** — squads, fixtures, results, standings, refreshed during the tournament.
- **Event-level history** — enough granularity (shots with xG, lineups) to power the same analytics for past tournaments.
- **Affordability** — this is a portfolio-grade build, not a funded product; recurring cost must be near-zero to low.
- **Licensing clarity** — terms must permit this use without bespoke negotiation.

The tension: rich historical event data and a cheap live feed are generally offered by different providers, with different schemas, granularity and licensing.

4. Goals & Constraints

4.1 Goals

- Real data on every analytical surface, for both live and historical tournaments.

- A clean abstraction so adding or swapping a provider never touches the engine, pages or AI layer.
- Predictable, low recurring cost.

4.2 Non-goals

- Broadcast-grade tracking data (player coordinates) — out of scope and out of budget.
- Real-time second-by-second event streams — periodic refresh is sufficient for the product's cadence.

5. Options Considered

5.1 Option A — StatsBomb open data only

What: use StatsBomb's free, event-level open data exclusively.

- **Pros** — free; genuinely rich (shot-level with xG, lineups, managers); clean licensing for the open set.
- **Cons** — it is historical. It does not cover the live 2026 tournament, which is the product's headline feature.

5.2 Option B — API-Football only

What: use API-Football for everything, live and historical.

- **Pros** — live 2026 coverage; broad competition catalog; affordable.
- **Cons** — shallower event granularity than StatsBomb for historical depth (e.g. shot-level xG, detailed positions); paying for history that is available free elsewhere.

5.3 Option C — Web scraping

What: scrape public sites for both live and historical data.

- **Pros** — nominally free; flexible coverage.
- **Cons** — brittle (markup changes break ingestion); ambiguous-to-prohibited licensing; high maintenance; ethically and legally the weakest option. Rejected on principle.

5.4 Option D — Hybrid (recommended)

What: StatsBomb open data for historical tournaments; API-Football for the live 2026 tournament; both normalized to one snapshot type.

- **Pros** — best data for each job: free, rich history and affordable, real-time live; each provider used to its strength.
- **Cons** — two schemas to map and two failure profiles to handle — absorbed entirely inside the adapter layer.

6. Comparison

Criterion	A · StatsBomb	B · API-Football	D · Hybrid
Live 2026	No	Yes	Yes
Historical depth	Excellent	Moderate	Excellent
Event granularity	Shot-level + xG	Summary-level	Best of both
Recurring cost	Free	~$19/mo	~$19/mo
Licensing	Clear (open set)	Clear (commercial)	Clear
Integration effort	Low	Low	Moderate

7. Recommendation

> **Decision** Adopt Option D (Hybrid). Use StatsBomb open data for historical tournaments and API-Football for the live 2026 tournament, behind a single adapter interface that emits the shared DatasetSnapshot.

The hybrid is the only option that satisfies the headline requirement (live 2026) **and** the depth requirement (event-level history) at near-minimum cost. Its single real drawback — two integrations — is contained: both adapters already target the same snapshot type, so the extra schema is a localized mapping problem, not a system-wide one.

8. Design Implications

- **Adapter interface** — each source implements one function returning a DatasetSnapshot. The engine, pages and AI layer remain unaware of the source.
- **Pre-computation** — StatsBomb history is fetched once and cached as static JSON (including group inference and manager extraction), so historical tournaments load instantly and offline.
- **Live path** — the API-Football adapter loads in the background at startup with paced, retried requests, and is keyed per-tournament in the cache.
- **Switchability** — because both resolve to the same type, a single control re-points the entire app between live and historical datasets.

9. Risks & Mitigations

Risk	Mitigation
API-Football rate-limits during burst load	Batched, paced requests with retry/backoff on HTTP- and body-level limits
StatsBomb open-data coverage gaps	Per-surface graceful degradation; never block a page on one missing entity

Risk	Mitigation
Provider schema change	Isolated in the adapter; the snapshot contract shields everything downstream
Cost creep	Single low-tier subscription; history is free and cached

10. Cost Analysis

Historical data (StatsBomb open set) is **free**. Live data (API-Football) is a single low-tier plan at roughly **$19/month**. There is no database or compute bill beyond hosting, because normalization and modeling run in-process. Total recurring cost is therefore dominated by one ~$19/month line item — proportionate to a portfolio-grade product and trivially cancellable out of season.

11. Rollout

1. Ship the StatsBomb adapter and pre-compute the historical tournaments (2018, 2022 men's; 2019, 2023 women's).
2. Ship the API-Football adapter behind a feature/source switch, defaulting to live when a key is present.
3. Verify parity: run the full data sweep across all datasets to confirm complete coverage per team.
4. Make the tournament selector the single switch across the whole app.

12. Decision & Sign-off

Role	Name	Decision
Author	Tobi Smith-Kayode	Proposed — Hybrid (Option D)
Decision owner	Tobi Smith-Kayode	Accepted
Date	June 2026	—